

Computational Fluid Dynamics

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Exercise 1.

Let us consider a fluid with kinematic viscosity $\nu = 0.01$. We want to simulate the viscous flow induced in a square cavity by the oscillating top boundary. The problem is governed by the time-dependent incompressible Navier-Stokes equations on the unit square domain $\Omega = [0, 1]^2$:

$$\left\{ \begin{array}{ll} \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \operatorname{div} \sigma = \mathbf{0} & \text{in } \Omega, \quad t \in (0, 1] \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \quad t \in (0, 1], \\ \mathbf{u} = \mathbf{u}_0 & \text{in } \Omega, \quad t = 0 \\ \mathbf{u} = [\sin(4\pi t), 0]^T & \text{on } \Gamma_{\text{top}} = \{(x, y), 0 \leq x \leq 1, y = 1\}, \quad t \in (0, 1], \\ \mathbf{u} = [0, 0]^T & \text{on } \partial\Omega \setminus \Gamma_{\text{top}}, \quad t \in (0, 1], \end{array} \right. \quad (1)$$

with $\sigma = \nu (\nabla \mathbf{u} + \nabla \mathbf{u}^T) - pI$.

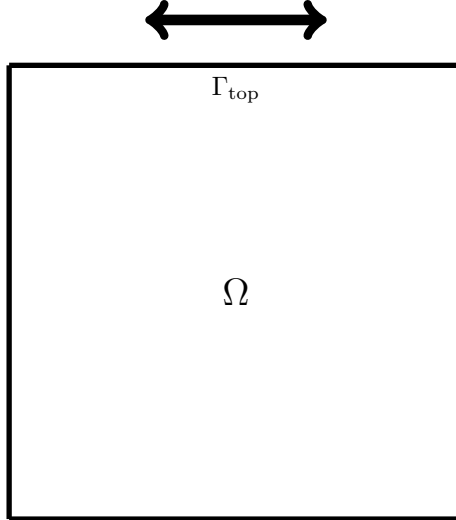


Figure 1: Computational domain

- a) Write the weak formulation of Problem (1).
- b) Introduce a time-discretization based on the backward Euler scheme with a semi-implicit treatment of the convective term.
- c) Based on the template `exam.edp`, write a `FreeFem++` script for the solution of Problem (1) using $\mathbb{P}_2 - \mathbb{P}_1$ finite elements for velocity and pressure, respectively. Consider the Cahouet-Chabard preconditioner for the solution of the linear system to be solved at each time iteration.
- d) Compute the solution considering an uniform grid with $h=0.1, 0.05, 0.025$ and time step $\Delta t=0.02$. Based on the numerical results, discuss the optimality of the preconditioner with respect to the space resolution.
- e) Compute the tangential force acting on the oscillating plate using the grid with $h=0.025$ and considering different time steps $\Delta t = 0.08, 0.04, 0.02$. For each of the 3 different time steps, report the maximum value of the tangential force in the time interval $[0, 1]$, as well as the final value at time $t = 1$.